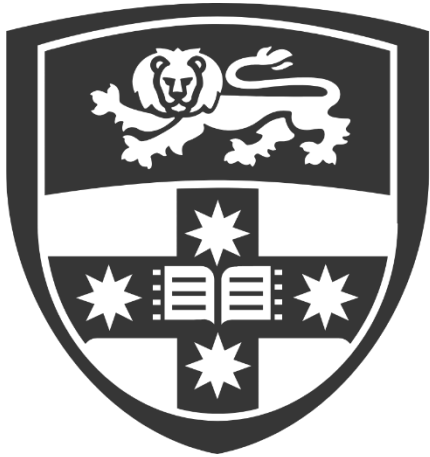




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Ultrasonic Crack Detection and Localisation in Steel Using Digital Signal Processing and Machine Learning

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1. Project Overview

This project proposes a signal-processing system for detecting and localising cracks and related flaws in steel from ultrasonic inspection data. Ultrasonic non-destructive testing is widely used because internal discontinuities reflect acoustic energy and can therefore be identified without damaging the component. The project will first establish a conventional digital signal processing (DSP) baseline, then evaluate whether machine-learning methods provide a measurable improvement in defect detection. The central research question is: how much does machine learning improve ultrasonic flaw detection over an interpretable DSP baseline, particularly as signal quality deteriorates?

2. Background and Motivation

Conventional automated ultrasonic inspection often relies on amplitude thresholds and engineered rules, while difficult weld inspections still require substantial human interpretation. Recent research has shown that convolutional neural networks can learn useful representations directly from phased-array ultrasonic B-scans, but representative flaw data remain scarce [1]. Koskinen et al. further demonstrated that the type and diversity of training flaws materially affect generalisation to unseen flaws [2]. This motivates a comparative approach rather than using a neural network as a black box.

The project is also well suited to advanced signal processing because Full Matrix Capture (FMC) records the time-domain response for every transmit-receive element pair. These data can be post-processed using delay-and-sum beamforming and the Total Focusing Method (TFM) to focus an image at each point in the inspection region [3], [4]. The combination of interpretable DSP, ultrasonic imaging and supervised learning provides technical depth while remaining feasible using public datasets.

3. Proposed Methodology

3.1 Data and Tools

Implementation will use Python with NumPy/SciPy for DSP, scikit-learn for classical machine learning, and PyTorch for the CNN. The primary labelled dataset will be NDT_ML_Flaw, which provides approximately 17,000 ultrasonic images across scanned and simulated flaw batches, with roughly balanced flaw/no-flaw examples and metadata including flaw depth, scan-axis location, size and type [2]. The ML-NDT dataset will be used as a secondary dataset; it contains 20,000 augmented phased-array ultrasonic samples derived from measured thermal-fatigue crack signals [1]. Separate Strathclyde FMC datasets containing known reflectors and cracks will be used for raw-signal DSP and localisation experiments.

3.2 Processing and Models



Three detection approaches will be compared on a common held-out test set. First, a conventional DSP baseline will use normalisation, denoising, time/spatial gating, envelope or amplitude processing, energy and peak features, followed by rule-based threshold detection. Second, the same processed signals will provide engineered features such as peak amplitude, RMS level, energy, peak position and spatial statistics to a support vector machine (SVM). Third, a compact convolutional neural network (CNN) will learn directly from ultrasonic B-scan images. This structure allows a controlled comparison between deterministic DSP, DSP-assisted ML and end-to-end feature learning.

For the FMC experiments, time-of-flight information will be combined across transmit-receive pairs using TFM beamforming to reconstruct the inspection region. Known flaw positions will permit localisation error to be calculated as $|d_{\text{est}} - d_{\text{true}}|$. Robustness will be evaluated by adding controlled noise to held-out test data at several signal-to-noise ratios (for example 20, 15, 10, 5 and 0 dB) and re-measuring performance. Dataset splits will be grouped by source flaw or batch wherever metadata permits, preventing augmented versions of the same source flaw from leaking across training and test sets.

3.3 Evaluation Protocol

To ensure that any improvement is attributable to the method rather than the data split, the DSP baseline, SVM and CNN will be evaluated using the same held-out examples. Model selection will be performed only on training/validation data, while final metrics will be reported once on the test set. Where possible, source flaws or augmentation batches will be kept within a single split to reduce data leakage. In addition to classification metrics, the

project will examine computational cost and robustness so that higher accuracy can be weighed against model complexity. This is particularly relevant for future real-time or embedded inspection systems.

3.4 Feasibility and Scope Control

The project is intentionally staged so that a complete result does not depend on deep-learning success or custom hardware. The minimum viable outcome is the DSP detector plus quantitative FMC localisation. The SVM comparison is the main machine-learning extension, while the CNN provides an additional end-to-end benchmark. No physical ultrasonic acquisition system will be required because all core experiments use public measured or augmented datasets. Equivalent crack-size estimation will only be attempted after the detection, localisation and robustness experiments are complete.

4. Expected Outcomes

The expected deliverable is a reproducible software prototype and GitHub repository implementing the complete ultrasonic analysis pipeline. Core outputs will include visualisations of representative A-scans/B-scans and TFM reconstructions, a DSP-based detector, an SVM using engineered features, and a compact CNN classifier. Detection performance will be reported using accuracy, precision, recall, F1 score, confusion matrices and ROC-AUC where appropriate. Localisation experiments will report absolute depth/location error, while the noise study will show performance versus SNR. If time permits, equivalent flaw-size estimation will be explored as a stretch objective rather than a required deliverable.

5. Timeline (Weeks 1-13)

Weeks	Planned Work
1-2	Finalise topic, repository and proposal.
3-5	Literature review; acquire and inspect datasets; implement data loaders and visualisation.
6-7	Develop DSP baseline and FMC/TFM localisation pipeline.
8-9	Extract DSP features; train and evaluate SVM; establish leakage-safe data splits.
10-11	Train compact CNN; compare DSP, SVM and CNN; conduct SNR robustness experiments.
12-13	Optimise and finalise evaluation; prepare report, GitHub documentation and demonstration.

6. References

- [1] I. Virkkunen, T. Koskinen, O. Jessen-Juhler, and J. Rinta-aho, "Augmented Ultrasonic Data for Machine Learning," *Journal of Nondestructive Evaluation*, vol. 40, art. 4, 2021. doi: 10.1007/s10921-020-00739-5.
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- [3] C. Holmes, B. W. Drinkwater, and P. D. Wilcox, "Post-processing of the full matrix of ultrasonic transmit-receive array data for non-destructive evaluation," *NDT & E International*, vol. 38, no. 8, pp. 701-711, 2005. doi: 10.1016/j.ndteint.2005.04.002.
- [4] C. Holmes, B. W. Drinkwater, and P. D. Wilcox, "The post-processing of ultrasonic array data using the total focusing method," *Insight - Non-Destructive Testing and Condition Monitoring*, vol. 46, no. 11, pp. 677-680, 2004. doi: 10.1784/insi.46.11.677.52285.

Primary public data resources: *NDT_ML_Flaw* (koomas/NDT_ML_Flaw), *ML-NDT* (iikka-v/ML-NDT), and *University of Strathclyde FMC datasets*.